

Strategic Human Resource Management
Analyzing Diversity: A Comparison of Northrop Grumman and Thales
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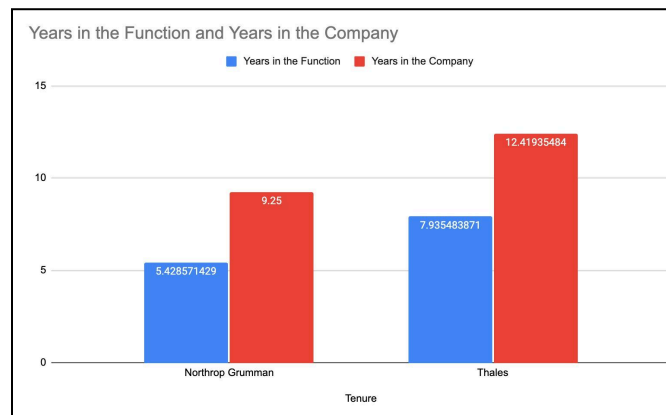
Introduction

This research aims to compare diversity in the governance structures of Northrop Grumman and Thales. The objective is to explore how diversity in key areas impacts organizational decision-making, leadership, and employee outcomes, offering insights into the significance of inclusive governance in global corporations. To give an introduction to the two companies chosen for my comparative analysis, **Northrop Grumman Corporation** is a leading global aerospace and defense technology company headquartered in Falls Church, Virginia, USA. Our second company is the **Thales Group**, headquartered in Paris, France, is a multinational company specializing in aerospace, defense, security, and digital technologies.

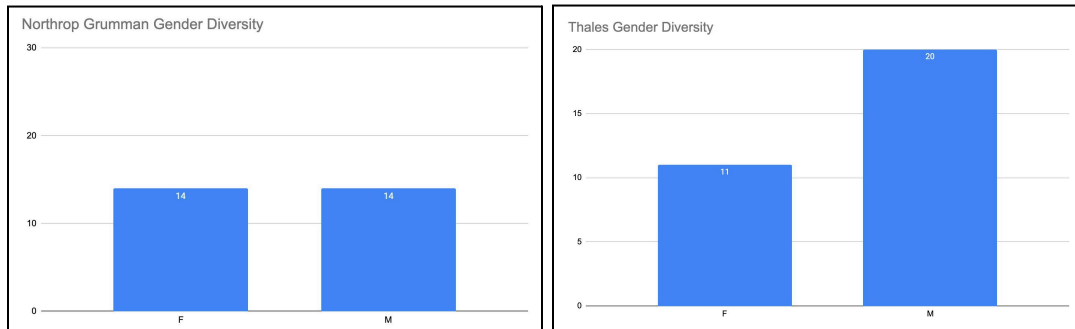
The approach we take in this analysis is to take several features of the employees in the companies, compiled as a dataset and perform some exploratory data analysis on them. This method enables us to not only view our dataset at a glance, but also helps us to give a strong statistical basis to all comments that we make about the company's diversity.

Analysis and Findings

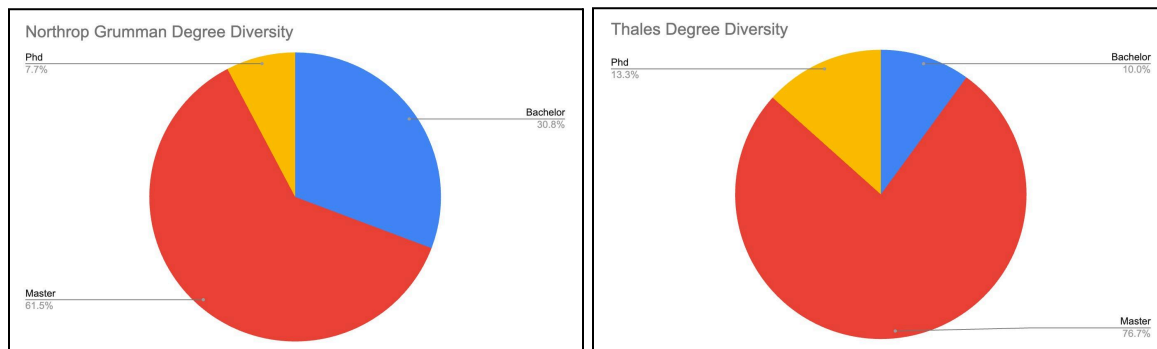
Each entry includes the *Company*, *Corporate Governance Body*, *Name*, *Function*, *Sex*, *Nationality*, *Year of Birth*, *Age (calculated for 2024)*, *Highest Degree*, *Educational Field*, *Institution*, *Years in the Function*, and *Years in the Company*. Let's take a look at some visualisations to help us compare the diversity within the two companies. For **Northrop Grumman** we have **28 entries** and for **Thales**, **31 entries**. While the difference in the dataset size is minimal, it suggests that the representation of roles or individuals within Thales is marginally broader.



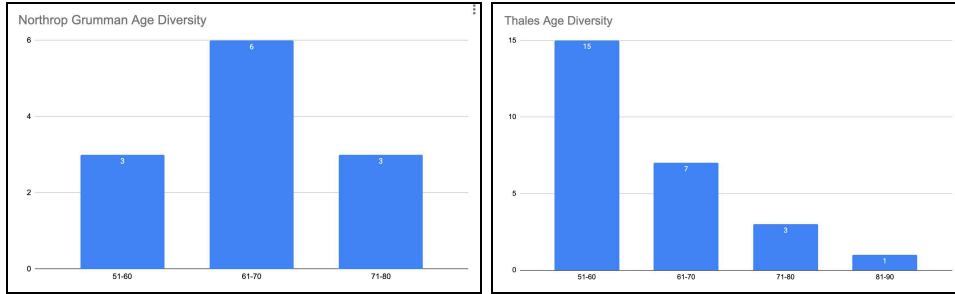
The first bar chart compares **years in function** and **years in the company** for Northrop Grumman and Thales. Employees at Thales have a longer average tenure in their current roles (**7.93 years**) compared to Northrop Grumman (**5.43 years**). In terms of overall tenure, Thales employees also stay with the company longer, averaging **12.42 years**, compared to **9.25 years** at Northrop Grumman. The wider gap between tenure in function and company at Thales points to employees staying within the organization but moving between roles less frequently. In contrast, Northrop Grumman's shorter tenures may reflect more dynamic role rotations or higher turnover



At **Northrop Grumman**, the gender distribution is balanced, with an equal number of male and female employees (14 each) in the analyzed dataset. In contrast, **Thales** shows a noticeable disparity, with 20 males compared to 11 females. While Northrop Grumman exhibits gender equality, Thales may benefit from initiatives to promote women into leadership roles.



At **Northrop Grumman**, the majority of employees (61.5%) hold a Master's degree, while Bachelor's degree holders account for 30.8%, and PhDs represent only 7.7%. At **Thales**, the preference for Master's degrees is even more pronounced, with 76.7% of employees holding this qualification. Bachelor's degrees account for 10%, while PhDs represent 13.3%, nearly double the proportion at Northrop Grumman. The comparison highlights a difference in educational priorities between the companies. While both prioritize Master's degrees, Thales demonstrates a stronger emphasis on PhDs, potentially reflecting its focus on innovation and research.



At **Northrop Grumman**, the majority of employees fall within the **61–70 age range** (6 individuals), while the **51–60** and **71–80 age ranges** each account for 3 employees. At **Thales**, there is a stronger emphasis on younger age groups, with the **51–60 age range** dominating (15 individuals). The **61–70 age range** includes 7 employees, while older groups, such as **71–80** and **81–90**, are sparsely represented, with 3 and 1 employees, respectively. These trends indicate that Thales might have a younger workforce in governance roles compared to Northrop Grumman, which appears to favor more experienced professionals. The missing age data in both datasets may limit the completeness of the analysis.

Impact of Sociological Features

To examine sociological impacts on careers, we use hypothesis testing in order to check if each of the sociological features have an impact on certain ‘career’ features such as the tenure. We perform these testings on the whole dataset, grouped by our two companies: Northrop Grumman and Thales.

I. Gender and Tenure

H_0 : There is no significant difference in tenure (Years in Function) between males and females.

H_A : There is a significant difference in tenure (Years in Function) between males and females.

The p-values for the Two-Sample T-Test assessing differences in tenure between genders are 0.867 for Northrop Grumman and 0.514 for Thales. Both values are above the conventional significance level (0.05), indicating that there is insufficient evidence to reject the null hypothesis in either case.

II. Educational Fields and Governance Roles

H_0 : Educational fields are independent of governance roles.

H_A : Educational fields are not independent of governance roles.

The p-values for the Chi-Square Test evaluating the independence of educational fields and governance roles are 0.595 for Northrop Grumman and 0.453 for Thales. These results indicate no significant relationship between educational fields and governance roles for either company. However, Thales exhibits a slightly lower p-value, suggesting a marginally closer association between the variables compared to Northrop Grumman, though still not statistically significant.

III. Age and Tenure

H_0 : There is no correlation between age and tenure (Years in Function).

H_A : There is a significant correlation between age and tenure (Years in Function).

For the correlation test between age and tenure, the p-value is 0.08 for Northrop Grumman. Conversely, Thales exhibited a significant correlation (p-value of 0.008), indicating that older employees tend to have longer tenure in the company. Northrop Grumman's p-value suggests marginal evidence against the null hypothesis; it does not reach significance while for Thales, we can reject the null hypothesis. This difference suggests that Thales may place greater emphasis on accumulated experience over time.

Conclusion

When it comes to diversity in corporate governance, Northrop Grumman and Thales are telling two distinct yet complementary stories. Northrop Grumman stands out with its gender-balanced leadership and emphasis on dynamic role transitions, fostering adaptability and innovation. In contrast, Thales has a highly educated workforce, favoring PhDs, and a preference for retaining experienced professionals in long-term roles, reflecting its focus on stability and expertise.

Yet, the numbers reveal interesting nuances about how sociological features shape careers. For instance, gender does not significantly affect tenure at either company, suggesting equality in role longevity once leadership is attained. However, Thales' stronger correlation between age and tenure hints at a workforce where experience is more tightly linked to career progression, while Northrop Grumman's looser relationship suggests more flexibility in leadership pathways. Thales' French roots might influence its slightly younger leadership and education-heavy approach, while Northrop Grumman's American base might drive its balance of innovation and inclusion.

The hypothesis testing demonstrates varying patterns across the examined relationships for Northrop Grumman and Thales. While the results do not show significant gender-based or educational disparities in tenure and governance roles, the significant age-tenure correlation at Thales highlights a nuanced difference in how age and experience influence workforce dynamics between the two organizations.

Appendix: [GitHub Link](#) with all necessary documents.